

OCR A A-Level Biology — Practical Activity Groups (PAGs)

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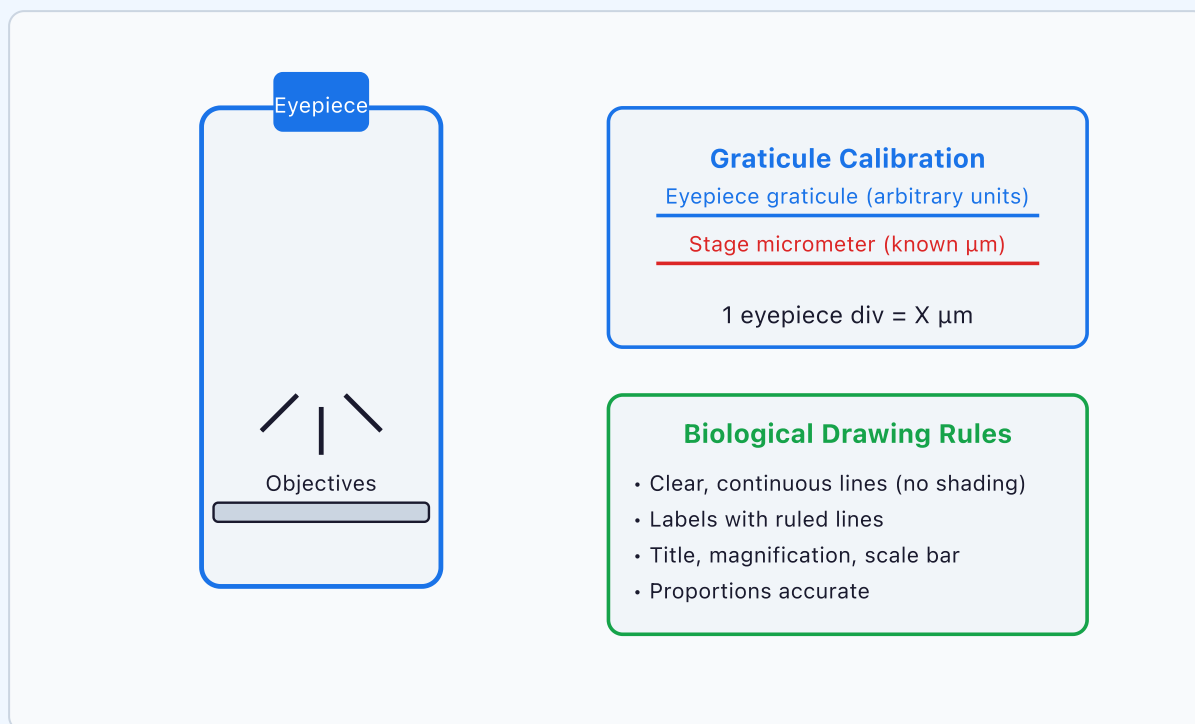
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Also on this page (not PAGs):

- [Supporting practical: Factors Affecting Respiration Rate](#)
- [Supporting practical: Factors Affecting Photosynthesis](#)

The OCR Practical Endorsement requires at least 12 practical activities covering PAG1 to PAG12. Each PAG is defined by the techniques and skills it must cover; the specification gives one example activity for each, and your school may use different activities. The practicals below are typical examples for each group.

PAG1: Microscopy



Apparatus

- Optical microscope, eyepiece graticule, stage micrometer
- Prepared slides or fresh material (e.g. onion epidermis)
- Microscope slides, coverslips, stains (iodine, methylene blue)
- Pencil, ruler, plain paper

Method

1. **Setting up:** Place slide on stage. Start with the lowest power objective (x4). Focus using coarse then fine focus.
2. **Calibration:** Place stage micrometer on stage. Align eyepiece graticule with stage micrometer. Count how many graticule divisions span a known number of micrometer divisions. Calculate: 1 eyepiece division = X μm .
3. **Measuring cells:** Replace micrometer with specimen. Count how many graticule divisions span a cell. Multiply by calibration factor.

4. **Drawing:** Produce a labelled biological drawing with clear single lines, no shading, correct proportions, title, and magnification.
5. **Calculating magnification:** Magnification = Image size / Actual size. Measure the drawing and the actual specimen (using calibrated graticule). Always show your working and include units.

Variables

Variable Type	Details
Independent	Type of tissue / cell being examined
Dependent	Cell size measurements (length, width, area)
Controlled	Magnification used, calibration factor, staining protocol, microscope settings

Key Calculations

Calibrating the eyepiece graticule: If 50 eyepiece divisions align with 10 stage micrometer divisions, and each stage micrometer division = 10 μm , then 50 eyepiece divisions = 100 μm , so 1 eyepiece division = 2 μm . This calibration must be repeated for each objective lens, as the calibration changes with magnification.

Calculating actual cell size: If a cell spans 25 eyepiece divisions and 1 eyepiece division = 2 μm , then cell size = $25 \times 2 = 50 \mu\text{m}$.

Safety

- Handle slides carefully — glass breakage risk.
- Stains may irritate skin — wash hands after use.
- Carry microscope with two hands.

Analysis

- Magnification = image size / actual size.
- Actual size = image size / magnification.
- Include a scale bar on drawings to show the actual size of structures.

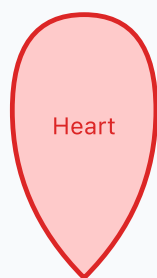
Sources of Error

Error	Effect	Improvement
Graticule not calibrated for each objective	Incorrect size measurements	Recalibrate when changing magnification
Parallax error reading scale	Inaccurate measurements	Ensure eye is directly over graticule
Air bubbles under coverslip	Distorts image	Lower coverslip at 45°; use correct volume of liquid
Drawing not to scale	Misleading representation	Use graph paper; calculate magnification

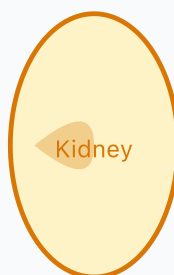
Exam Tip: The formula triangle: *Magnification = Image size / Actual size*. Recalibrate the eyepiece graticule every time you change objective lens. Include units (μm) and a scale bar on all drawings. Total magnification = eyepiece $\times$ objective.



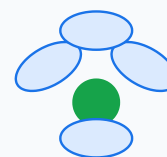
PAG2: Dissection



Identify: ventricles,
atria, valves, vessels



Identify: cortex,
medulla, pelvis, ureter



Identify: sepals, petals,
stamens, carpel, ovary

Tools: scalpel, scissors, forceps, pins, dissecting board, hand lens

Apparatus

- Specimens: heart, kidney, and/or flower (fresh or preserved)
- Dissecting board, pins, scalpel, scissors, forceps, mounted needle
- Hand lens or dissecting microscope, gloves, eye protection

Method

1. **Heart:** Identify external features (coronary arteries, major vessels). Cut open to reveal chambers, valves, and wall thickness. Compare left and right ventricle walls.
2. **Kidney:** Cut longitudinally. Identify cortex, medulla (pyramids), pelvis, and ureter.
3. **Flower:** Remove and identify sepals, petals, stamens (anthers + filaments), and carpel (stigma, style, ovary). Cut ovary to reveal ovules.
4. Produce labelled biological drawings of each dissection.

Safety

- Wear gloves with biological specimens. Wash hands thoroughly after.
- Sharp tools — cut away from body on dissecting board.
- Dispose of biological material in designated waste.

Analysis

- Heart: left ventricle wall is thicker (pumps blood to whole body at higher pressure). Valves prevent backflow.
- Kidney: cortex contains glomeruli; medulla contains loops of Henle and collecting ducts.
- Flower: relate structures to pollination method (insect vs wind pollinated).

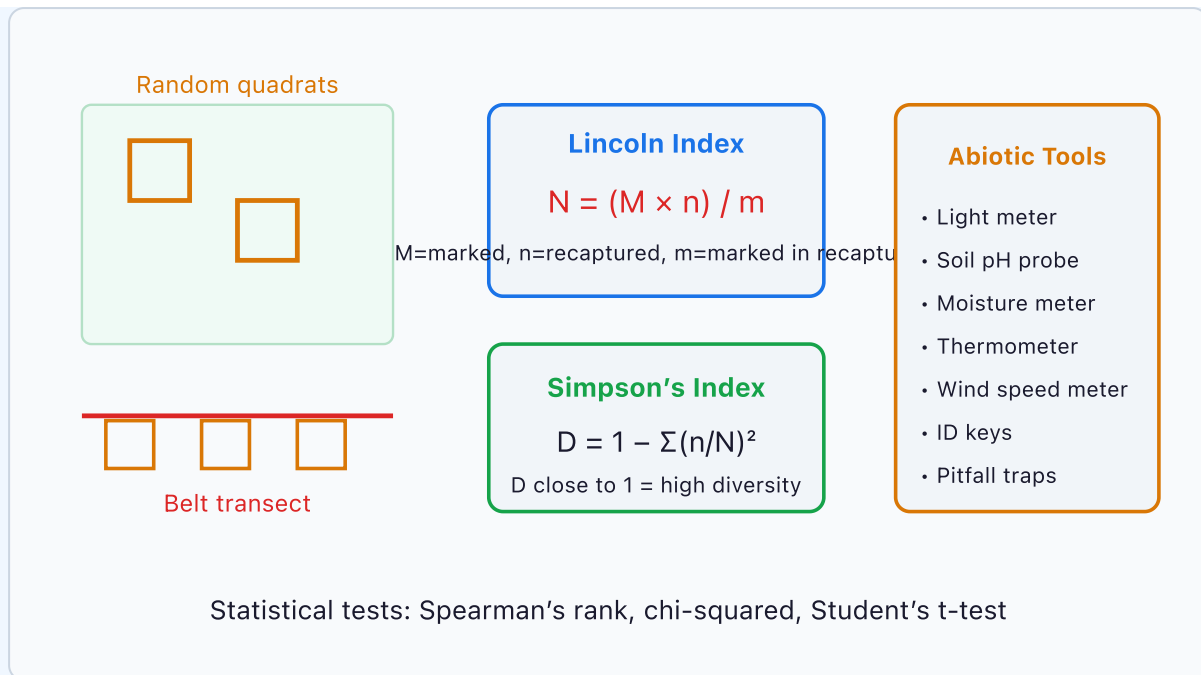
Sources of Error

Error	Effect	Improvement
Structures damaged during dissection	Difficult to identify	Use fine instruments; cut carefully
Specimen dried out	Structures shrink	Keep moist; work quickly
Incorrect identification	Inaccurate drawings/labels	Use reference diagrams; check with teacher

Exam Tip: For the heart: the left ventricle has a thicker muscular wall because it needs to generate higher pressure to pump blood around the entire body (systemic circulation). The right ventricle only pumps to the lungs (pulmonary circulation).



PAG3: Sampling Techniques



Apparatus

- Quadrats, tape measure, random number table
- Pitfall traps, sweep nets, identification keys
- Abiotic tools: light meter, soil pH probe, moisture meter, thermometer
- Calculator, recording sheets

Method

1. **Random sampling:** Generate random coordinates. Place quadrats. Record species and abundance in each.
2. **Systematic sampling:** Lay transect line. Place quadrats at intervals. Record species and abiotic factors at each point.
3. **Mark-release-recapture:** Capture, mark, release. Recapture after mixing. Use Lincoln index: $N = Mn/m$.
4. Calculate Simpson's Diversity Index for each area sampled.

Safety

- Fieldwork risk assessment. Wash hands after soil/animal contact.
- Be aware of hazardous organisms (nettles, ticks).

Analysis

- Species density = mean count per quadrat / quadrat area.
- Simpson's $D = 1 - \sum(n/N)^2$. Compare D values for different habitats.
- Spearman's rank to test correlation between distribution and abiotic factor.

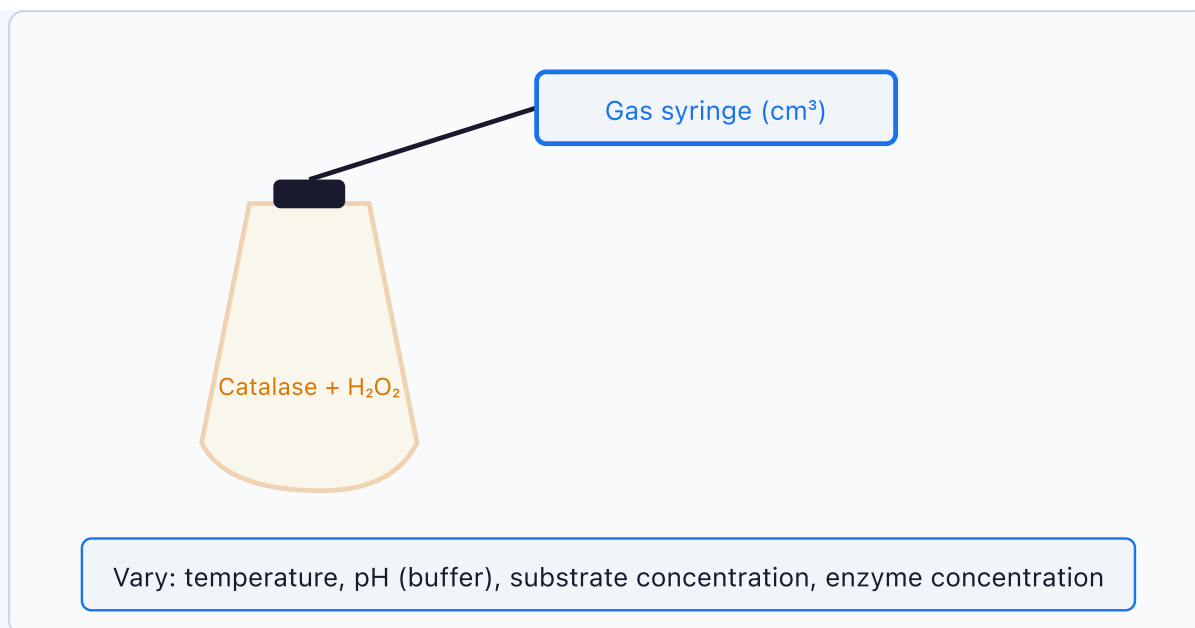
Sources of Error

Error	Effect	Improvement
Non-random quadrat placement	Biased sampling	Use random number coordinates
Too few quadrats	Unrepresentative sample	At least 10 quadrats per area
Marks affect survival (MRR)	Population estimate biased	Use small, inconspicuous marks
Misidentification	Incorrect diversity calculations	Use identification keys; photograph specimens

Exam Tip: Know the assumptions of mark-release-recapture: no births/deaths/migration between captures, marks do not affect survival, random mixing occurs. For Simpson's index, always show full working. Spearman's rank tests for correlation between two variables.



PAG4: Rates of Enzyme-Controlled Reactions



Apparatus

- Catalase source (potato discs), H₂O₂ solutions
- Conical flask, bung, delivery tube, gas syringe
- Water baths, buffer solutions, stopwatch, thermometer

Method

1. Set up apparatus. Equilibrate enzyme and substrate at target condition (5 min).
2. Mix. Immediately fit bung. Record O₂ volume every 30 s for 5 min.
3. Vary one factor at a time (temperature, pH, substrate concentration).
4. 3 repeats per condition. Calculate initial rate from the steepest portion of the curve.

Safety

- H₂O₂ is an irritant — eye protection and gloves.
- Care with hot water baths.

Analysis

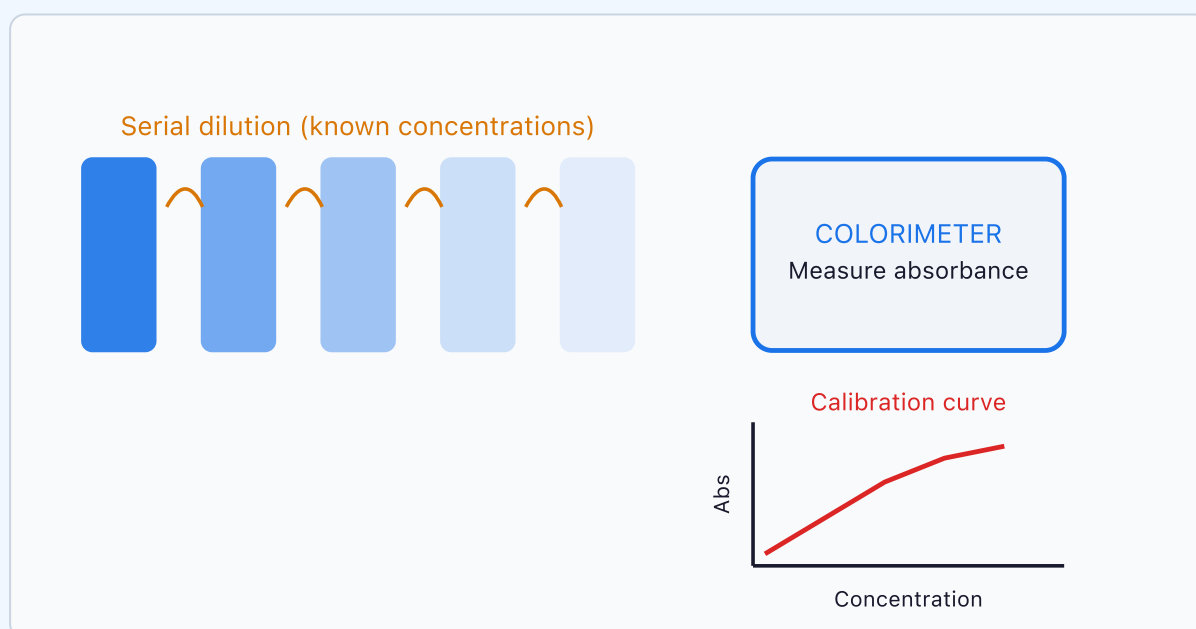
- Plot volume against time. Calculate initial rate = gradient of tangent at t=0.
- Plot rate vs variable: bell-shaped curve for temp/pH; saturation curve for substrate concentration.

Sources of Error

Error	Effect	Improvement
Gas leaks	Volume underestimated	Check seals; use Vaseline
Delay fitting bung	Gas lost at start	Practise technique; use side-arm flask
Variable enzyme amount	Unfair comparison	Use same mass of potato each time

Exam Tip: Always calculate the *initial rate* to avoid complications from substrate depletion. For temperature: explain the optimum using collision theory and denaturation. For substrate: explain the plateau using enzyme saturation (all active sites occupied).

PAG5: Colorimeter or Potometer



Apparatus

- Stock solution (e.g. glucose), distilled water

- Test tubes, graduated pipettes, Benedict's reagent
- Colorimeter, cuvettes, appropriate filter
- Water bath (for Benedict's test)

Method

1. Prepare serial dilution of known glucose concentrations.
2. Add Benedict's reagent to each and heat in water bath at 80 °C for 5 minutes.
3. Cool. Filter if precipitate is present (or use supernatant after settling).
4. Zero the colorimeter with a blank. Measure absorbance of each solution.
5. Plot calibration curve. Test unknown sample and read concentration from curve.

Safety

- Benedict's contains copper sulfate — irritant. Wear eye protection.
- Hot water bath — handle tubes with care.

Analysis

- For Benedict's quantitative: more glucose = more copper reduced = *lower* blue colour remaining = *lower* absorbance (at red filter).
- Or measure the absorbance of precipitate colour (red/orange) instead.
- Read unknown concentration from the calibration curve.

Sources of Error

Error	Effect	Improvement
Precipitate in cuvette	Scatters light, incorrect reading	Filter or centrifuge before reading
Air bubbles in cuvette	False absorbance reading	Tap cuvette gently; wipe with tissue
Wrong filter	Low sensitivity	Use complementary colour filter

Other PAG5 Activities

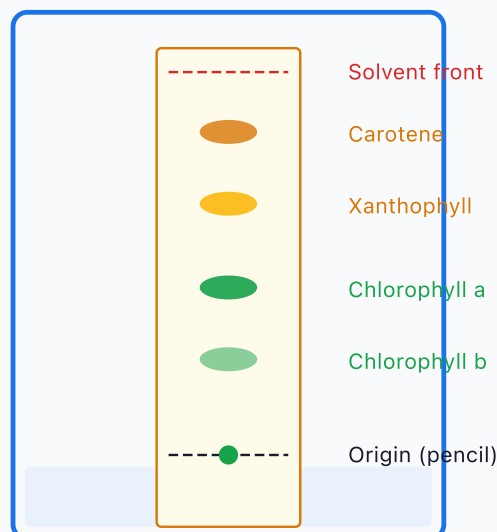
- **Membrane permeability (colorimeter):** Place equal-sized beetroot cubes (washed to remove pigment from cut cells) in water at a range of temperatures for a fixed time. Measure the absorbance of the surrounding liquid. Higher temperatures damage the phospholipid bilayer and denature membrane proteins, so more red betalain pigment leaks out and absorbance rises. Organic solvents such as ethanol have a similar effect.
- **Potometer:** Cut a leafy shoot under water and fit it into a potometer without air bubbles; seal joints. Introduce an air bubble and time its movement along the capillary. Rate of water uptake = distance moved $\times \pi r^2$ of the capillary $\div$ time. Vary light, air movement (fan), temperature or humidity. A potometer measures water *uptake*, which is close to, but not exactly, the rate of transpiration (some water is used in photosynthesis and to keep cells turgid).

Exam Tip: Semi-quantitative Benedict's: the colour change (blue $\rightarrow$ green $\rightarrow$ yellow $\rightarrow$ orange $\rightarrow$ brick red) indicates increasing glucose. For a quantitative result, use a colorimeter. The calibration curve converts absorbance to concentration objectively.



PAG6: Chromatography or Electrophoresis

$$R_f = \frac{\text{pigment dist.}}{\text{solvent dist.}}$$



Apparatus

- Fresh leaves, pestle and mortar, sand, propanone
- TLC plate or chromatography paper, pencil, ruler
- Chromatography jar with lid, capillary tube

Method

1. Grind leaves with sand and propanone. Filter extract.
2. Draw pencil origin line on TLC plate/paper. Apply extract spots using capillary tube. Dry between applications.
3. Place in jar with solvent below origin line. Seal.
4. Allow solvent to rise. Mark solvent front. Dry. Measure distances.
5. Calculate R_f for each pigment.

Safety

- Propanone is highly flammable — no flames; use fume cupboard.
- Avoid skin contact with solvents.

Analysis

- R_f = distance moved by pigment / distance moved by solvent.
- Order (highest R_f first): carotene > xanthophyll > chlorophyll a > chlorophyll b.
- Carotene is most soluble in non-polar solvent (most non-polar pigment).

Sources of Error

Error	Effect	Improvement
Origin below solvent line	Pigments dissolve off	Ensure solvent is below origin
Lid not sealed	Uneven solvent front	Use well-sealed container
Used pen for origin	Ink runs with solvent	Always use pencil

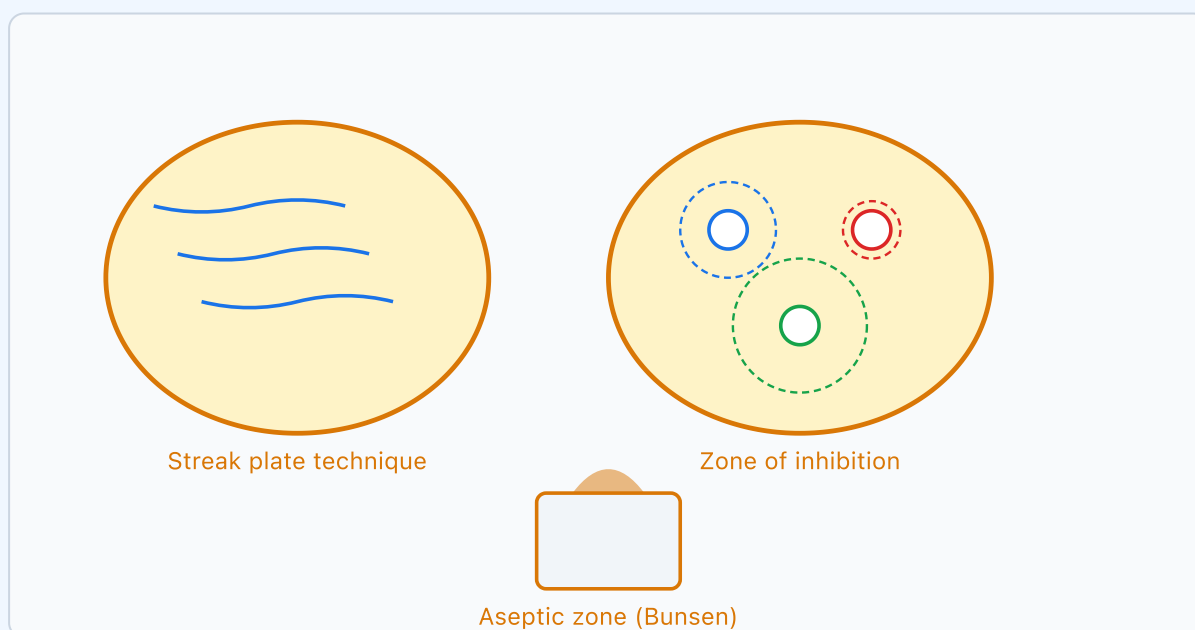
Other PAG6 Activities

- **Amino acids by paper chromatography:** Hydrolyse a protein into its amino acids, spot the hydrolysate and known amino acid standards on the origin line, and run in a suitable solvent. Amino acids are colourless, so spray the dried paper with ninhydrin (in a fume cupboard, with gloves) and warm it to reveal purple spots. Identify amino acids by comparing R_f values with the standards.
- **Gel electrophoresis:** DNA fragments (e.g. from restriction digestion) are loaded into wells at the cathode end of an agarose gel. DNA is negatively charged (phosphate groups), so it moves towards the anode; smaller fragments move further. Compare fragment lengths using a DNA ladder of known sizes.

Exam Tip: R_f values are specific to a given solvent at a given temperature. Carotene has the highest R_f because it is the most non-polar pigment and is most soluble in the non-polar running solvent. Chlorophyll b is the most polar and moves least.



PAG7: Microbiological Techniques



Apparatus

- Nutrient agar plates, bacterial broth culture
- Inoculating loop, Bunsen burner, antibiotic discs
- Sterile forceps, sterile spreader, marker pen, tape

Method

1. **Streak plate:** Flame loop, dip in culture, streak across plate in parallel lines. Turn plate 90°, flame loop, streak through previous lines. Repeat. This dilutes the culture to produce individual colonies.
2. **Zone of inhibition:** Spread bacteria evenly. Place antibiotic discs using flamed forceps. Seal with tape. Incubate upside down at $\leq 25^{\circ}\text{C}$ for 24–48 hours.
3. **Serial dilution:** Dilute broth culture in series (1 in 10 dilutions). Plate each dilution to count viable bacteria.

Safety

- Incubate $\leq 25^{\circ}\text{C}$ in schools. Do not open plates after incubation.
- Autoclave all materials before disposal.
- Aseptic technique at all times to prevent contamination.

Analysis

- Zone of inhibition: measure diameter, calculate πr^2 . Larger area = more effective antimicrobial.
- Streak plate: individual colonies can be subcultured for pure cultures.
- Serial dilution: count colonies $\times$ dilution factor = bacteria per cm^3 .

Sources of Error

Error	Effect	Improvement
Loop not flamed between streaks	No dilution of bacteria	Flame loop each time; allow to cool
Contamination	Unwanted colonies	Work near Bunsen; flame all equipment

Plates opened after incubation

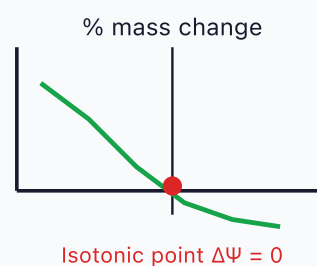
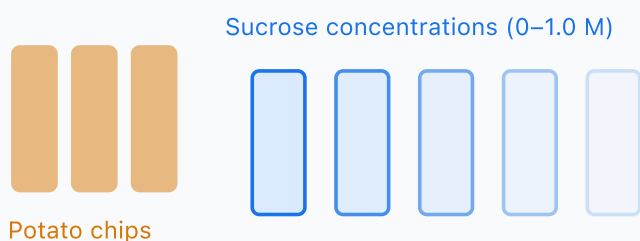
Exposure to
pathogens

Keep sealed; observe through lid

Exam Tip: Aseptic technique includes: working near a Bunsen (updraft prevents contamination), flaming inoculating loops and bottle necks, not breathing over plates, sealing with tape (but not completely — to prevent dangerous anaerobes growing). Incubation $\leq 25^\circ\text{C}$ prevents human pathogens growing.



PAG8: Transport In and Out of Cells



Apparatus

- Potato, cork borer, scalpel, ruler
- Sucrose solutions (0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm⁻³)
- Test tubes, balance (2 d.p.), paper towels, labels

Method

1. Cut uniform potato chips. Blot dry and weigh.

2. Place one chip in each sucrose concentration for 30–60 min.
3. Remove, blot dry, reweigh. Calculate % change in mass.
4. Repeat 3 times per concentration.

Safety

- Care with sharp tools. Solutions are non-hazardous.

Analysis

- Plot % mass change (y-axis) vs sucrose concentration (x-axis).
- Where the line crosses zero = water potential of potato cells equals the external solution.
- In dilute solutions: water enters by osmosis (mass increases). In concentrated: water leaves (mass decreases).

Sources of Error

Error	Effect	Improvement
Unequal chip sizes	Different SA:V ratios	Use same cork borer; cut to exact length
Inconsistent blotting	Variable surface water	Standardise blotting technique
Too short in solution	Equilibrium not reached	Leave for at least 30 min

Exam Tip: Use *percentage* change in mass to allow fair comparison between chips of slightly different starting masses. Water moves by osmosis from a region of higher water potential (Ψ) to lower water potential. At the isotonic point, there is no net movement of water.



PAG9: Qualitative Testing for Biological Molecules



Food Test Summary

Reducing sugar: Benedict's + heat

Non-reducing: HCl + neutralise + Benedict's

Starch: Iodine (brown → blue-black)

Protein: Biuret (blue → purple)

Lipid: Emulsion test (milky white)

Non-reducing sugar test: hydrolyse with HCl, neutralise with NaHCO₃, then repeat Benedict's test

Apparatus

- Test tubes, test tube holder, water bath (80 °C)
- Benedict's reagent, iodine solution, biuret reagent (NaOH + CuSO₄)
- Ethanol, dilute HCl, sodium hydrogen carbonate (NaHCO₃)
- Food samples: glucose solution, starch solution, egg albumin, vegetable oil, sucrose solution

Method

1. **Reducing sugars:** Add Benedict's to sample. Heat in water bath at 80 °C for 5 min. Blue → green → yellow → orange → brick red = positive.
2. **Non-reducing sugars:** If Benedict's is negative, add dilute HCl. Heat. Neutralise with NaHCO₃. Repeat Benedict's. Positive now = non-reducing sugar present.
3. **Starch:** Add iodine solution. Brown/yellow → blue-black = positive.
4. **Protein:** Add NaOH then CuSO₄ (biuret reagent). Blue → purple/lilac = positive.
5. **Lipids:** Dissolve sample in ethanol. Pour into water. White emulsion = positive.

Safety

- Benedict's and biuret contain copper sulfate — irritant.
- NaOH (in biuret) is corrosive — wear gloves and eye protection.

- Ethanol is flammable — no naked flames.
- HCl is corrosive — handle with care.

Analysis

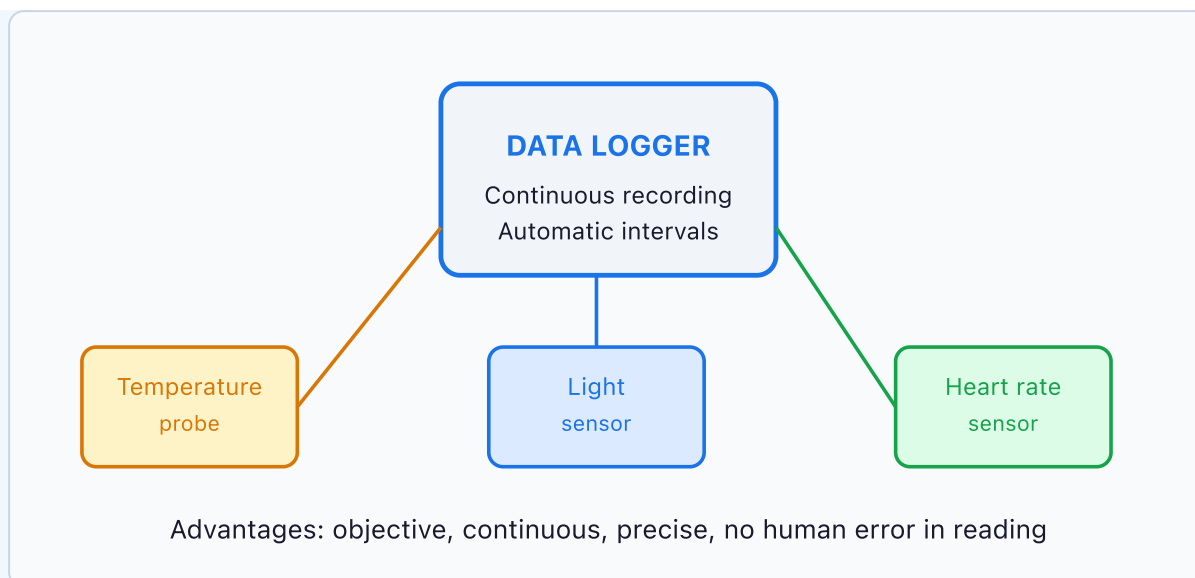
- These are qualitative tests (positive/negative). For quantitative results, use a colorimeter with calibration curve.
- Non-reducing sugar test: hydrolysis with HCl breaks the glycosidic bond (e.g. in sucrose), releasing glucose and fructose which then give a positive Benedict's result.

Sources of Error

Error	Effect	Improvement
Not heating long enough	Benedict's stays blue (false negative)	Heat for at least 5 minutes at 80 °C
Not neutralising after HCl	Acid destroys Benedict's reagent	Add NaHCO ₃ until pH neutral before testing
Cross-contamination of samples	False positives	Use clean test tubes and pipettes for each test

Exam Tip: For non-reducing sugars (e.g. sucrose): you must first hydrolyse with acid, then neutralise, then test with Benedict's. If you forget to neutralise, the acid will break down Benedict's reagent and give a false result. Always state the colour change as evidence.

PAG10: Investigation Using a Data Logger or Computer Modelling



Apparatus

- Data logger with appropriate sensor (temperature probe, light sensor, pH probe, heart rate monitor, O₂ sensor)
- Computer or display unit for data capture
- Appropriate experimental setup depending on investigation

Method

1. Choose an investigation (e.g. monitoring heart rate during exercise, temperature change during enzyme reaction, light intensity at different distances).
2. Connect the appropriate sensor to the data logger. Calibrate the sensor.
3. Set the recording interval and duration. Start recording.
4. Conduct the experiment. The data logger automatically records continuous data.
5. Export data to a computer for analysis and graph plotting.

Computer modelling alternative: Use molecular-visualisation software (e.g. RasMol or Jmol) to display a DNA or protein structure. Rotate the model and switch display styles to identify the sugar–phosphate backbone, complementary base pairing, the antiparallel strands and the double helix; for proteins, pick out α -helices, β -pleated sheets and the active site of an enzyme.

Safety

- Depends on specific investigation. Follow standard safety protocols.

- Keep electronic equipment away from water and heat sources.

Analysis

- Advantages of data logging: objective (no human judgement), continuous recording, precise readings, no parallax error, can record rapid changes.
- Export data to spreadsheet for statistical analysis and graph production.

Sources of Error

Error	Effect	Improvement
Sensor not calibrated	Systematic error in all readings	Calibrate against known standard before use
Recording interval too long	Misses rapid changes	Choose appropriate interval for the process
Sensor drift	Readings become inaccurate over time	Recalibrate periodically; use fresh sensors

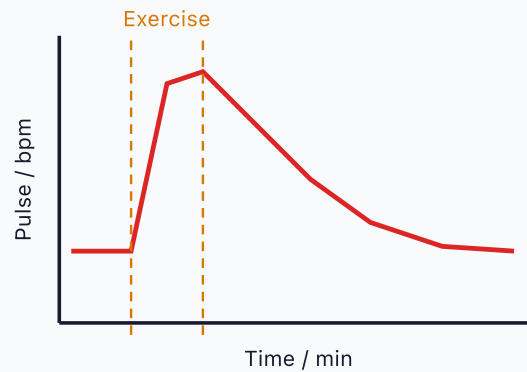
Exam Tip: Data loggers provide objective, continuous data that eliminates human error in reading instruments. They are particularly useful for measurements that change rapidly (e.g. heart rate) or need continuous monitoring (e.g. temperature over hours).



PAG11: Measuring Plant or Animal Responses

Measuring Pulse Rate

- Rest 5 min → resting rate
- Fixed exercise (e.g. step-ups)
- Measure every minute after
- Recovery time to resting rate



Other suitable responses

ion time • ventilation (spirometer) • phototropism / gravitropism in seedlings • invertebrate behaviour (choice chamber)

Apparatus

- Stopwatch, or heart-rate monitor / pulse sensor connected to a data logger
- Step or bench of fixed height, metronome to set the stepping pace
- Recording table; access to a seat for resting measurements

Method

1. Sit quietly for 5 minutes. Measure resting pulse at the radial (wrist) or carotid (neck) artery with two fingers, counting for 30 s and doubling (or read the monitor). Take several readings and calculate a mean.
2. Carry out a standard exercise at a fixed intensity for a fixed time (e.g. step-ups in time with a metronome for 3 minutes).
3. Measure pulse immediately after exercise and then every minute until it returns to the resting value. Record the recovery time.
4. Vary one factor (e.g. duration or intensity of exercise, or compare trained and untrained participants), keeping everything else the same.
5. Collect data from several participants to allow a statistical comparison.

Variables

Variable Type	Details
Independent	Intensity or duration of exercise (or fitness of participant)
Dependent	Pulse rate (beats min ⁻¹) and recovery time
Controlled	Type of exercise, step height, pace, resting period beforehand, body position when measuring, time since last meal/caffeine

Safety and Ethics

- Participants must give informed consent and be free to withdraw at any time.
- Screen out anyone with a heart condition, asthma or injury; stop if dizzy or unwell.
- Keep personal data anonymous. For plant or invertebrate work, handle organisms carefully and return animals to their habitat.

Analysis

- Plot pulse rate against time, marking the exercise period. Compare peak rate and recovery time between conditions.
- Exercise increases respiration in muscles, so CO₂ rises and blood pH falls; chemoreceptors detect this and the cardiovascular centre in the medulla increases impulses along the sympathetic nerve to the SAN, raising heart rate. Adrenaline also increases heart rate.
- After exercise, parasympathetic (vagus nerve) impulses slow the heart back to its resting rate.
- Use a Student's t-test to compare mean values for two groups (e.g. trained vs untrained).

Sources of Error

Error	Effect	Improvement
Counting pulse by hand over a short time	Low precision; beats missed	Count for longer or use a pulse sensor/data logger

Exercise intensity not constant	Unfair comparison	Use a metronome and fixed step height
Participants differ (age, fitness, sex)	Uncontrolled variables	Use large samples; match participants or use the same person
Delay before measuring after exercise	Peak rate underestimated	Measure immediately, ideally continuously

Exam Tip: PAG11 covers the safe and ethical use of organisms to measure responses and physiological functions, so ethics (consent, right to withdraw, minimising harm) is often credited. For heart rate control, name the pathway: chemoreceptors/baroreceptors → medulla (cardiovascular centre) → sympathetic or vagus nerve → SAN.

PAG12: Research Skills



Example: respiration rate of yeast (*Saccharomyces cerevisiae*)

- Yeast + glucose in buffer, water bath at set temperature
- Measure CO₂ (gas syringe / bubbles) or O₂ uptake (respirometer)
- Or: rate of decolourisation of methylene blue / DCPIP

What PAG12 Assesses

- Applying investigative approaches: turning a research question into a testable hypothesis and a workable plan.

- Using online and offline research (textbooks, journals, reputable websites) to inform the method and to compare results with published values.
- Correctly citing sources of information, e.g. in a consistent referencing style (author, year, title, publisher or URL, date accessed).

Apparatus (yeast respiration example)

- Dried or fresh yeast suspension, glucose (and other sugars if comparing substrates), buffer solution
- Boiling tubes or conical flask, bung with delivery tube and gas syringe (or respirometer)
- Thermostatically controlled water bath, thermometer, stopwatch, graduated pipettes

Method

1. Research the factors that affect respiration in yeast (e.g. temperature, substrate type or concentration, oxygen availability) and choose one as the independent variable. Record every source you use.
2. Write a hypothesis and a plan, including a risk assessment, the range of values to test and how other variables will be controlled.
3. Equilibrate yeast suspension and substrate solution separately in the water bath (about 10 minutes).
4. Mix, seal quickly and record the volume of CO₂ produced at regular intervals (or the distance moved by the fluid in a respirometer).
5. Repeat each condition at least three times; include a control (e.g. no substrate, or boiled yeast).
6. Write up the investigation, comparing your findings with the published sources and citing them in a reference list.

Variables

Variable Type	Details
Independent	e.g. temperature, or type/concentration of respiratory substrate

Dependent	Rate of CO ₂ production (cm ³ min ⁻¹) or O ₂ uptake
Controlled	Mass/concentration of yeast, volume and pH (buffer) of solution, temperature (if not the IV), time allowed to equilibrate

Safety

- Care with hot water baths and glassware.
- Yeast is low hazard, but wash hands and avoid sealing vessels completely without a gas outlet (pressure build-up).

Analysis

- Plot volume of CO₂ against time and use the initial gradient as the rate for each condition.
- Explain results using the stages of respiration: e.g. yeast respires glucose readily, but may respire some disaccharides only slowly or not at all if it lacks the enzyme to hydrolyse them.
- Compare your results with published data; suggest reasons for any differences and evaluate the reliability of your sources.

Sources of Error

Error	Effect	Improvement
Yeast suspension not mixed	Different amounts of yeast in each tube	Stir suspension before every sample
Gas lost before sealing	Rate underestimated	Seal immediately; use a side-arm flask
Unreliable or uncited sources	Weak justification of method and conclusions	Use peer-reviewed or reputable sources and reference them fully

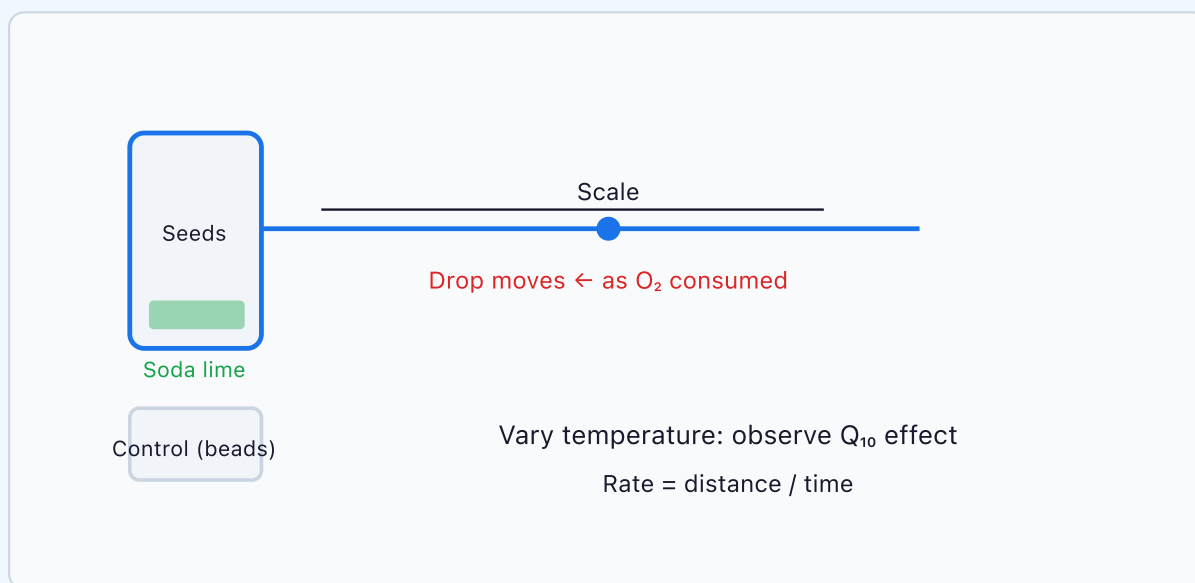
Exam Tip: Written papers can ask you to evaluate a method or a source: comment on whether a source is peer reviewed, how recent it is, sample size, and possible bias. When

planning, always state the independent, dependent and control variables, the number of repeats and how you would analyse the data.

Other Practical Techniques (not PAGs)

The two practicals below are **not** Practical Activity Groups in the OCR H420 specification, but both techniques appear in the Module 5 content (photosynthesis and respiration) and can be examined in the written papers. A respirometer investigation can also be used as the activity for PAG11 or PAG12.

Supporting Practical: Factors Affecting Respiration Rate



Apparatus

- Simple respirometer, germinating seeds (e.g. peas or mung beans)
- Glass beads (control), soda lime, cotton wool
- Capillary tube, coloured fluid, water bath, stopwatch

Method

1. Place seeds on soda lime (separated by cotton wool) in one tube. Control tube has glass beads on soda lime.
2. Seal tubes and connect to capillary. Place in water bath. Equilibrate 5 min.
3. Introduce fluid drop. Record distance moved per minute for 10 minutes.
4. Repeat at different temperatures (e.g. 15, 20, 25, 30, 35 °C).

Safety

- Soda lime is corrosive — wear gloves and goggles.

Analysis

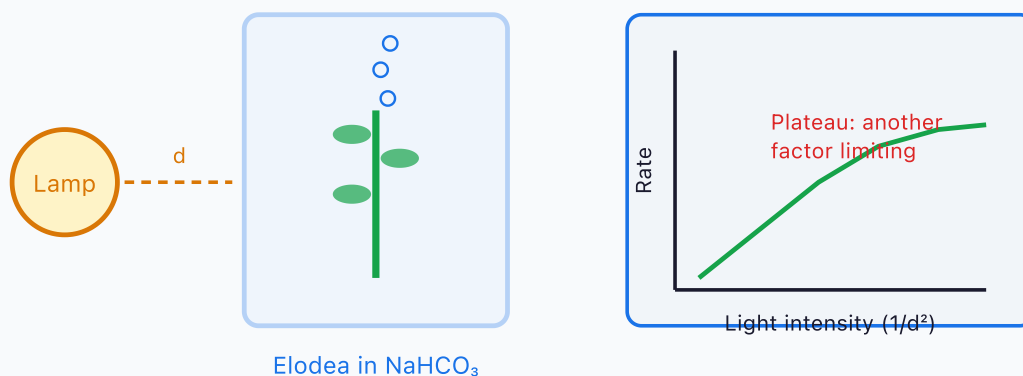
- Rate = distance moved / time. Plot rate vs temperature.
- Calculate Q_{10} = rate at (T+10) / rate at T. Expected ~2 for enzyme-catalysed reactions.
- Rate decreases above optimum due to enzyme denaturation.

Sources of Error

Error	Effect	Improvement
Leaks in apparatus	No fluid movement	Seal joints with Vaseline
Seeds at different germination stages	Variable respiration rates	Use seeds germinated for same duration
Soda lime exhausted	CO ₂ not absorbed	Use fresh soda lime

Exam Tip: Q_{10} = rate at (T+10) / rate at T. A Q_{10} of ~2 indicates the rate is controlled by enzyme-catalysed reactions. The control tube accounts for physical changes (gas expansion) due to temperature, not biological respiration.

Supporting Practical: Factors Affecting Photosynthesis



Count bubbles per minute at each distance. $I \propto 1/d^2$

Apparatus

- Elodea (pondweed), beaker, NaHCO₃ solution
- Bench lamp, ruler, stopwatch
- Thermometer, heat shield (beaker of water)

Method

1. Place Elodea in NaHCO₃ solution. Cut stem at an angle. Equilibrate for 2 min.
2. Position lamp at a set distance. Count bubbles per minute (3 repeats, take mean).
3. Move lamp to different distances (5, 10, 15, 20, 25 cm).
4. Light intensity $\propto 1/d^2$.

Safety

- Keep lamp away from water. Use heat shield between lamp and beaker.

Analysis

- Plot rate of bubble production against $1/d^2$ (light intensity).
- Initially proportional (light is limiting factor), then plateaus (CO₂ or temperature becomes limiting).
- NaHCO₃ provides CO₂ to prevent it being limiting.

Sources of Error

Error	Effect	Improvement
Unequal bubble sizes	Volume not proportional to count	Use gas syringe for volume
Heat from lamp	Temperature confounds results	Use heat shield; monitor with thermometer
Background light	Additional photosynthesis	Darken room
Plant condition varies	Different rates between replicates	Use same piece of Elodea; same condition throughout

Exam Tip: The law of limiting factors states that the rate of photosynthesis is limited by the factor in shortest supply. On the graph, the plateau shows where a different factor (e.g. CO₂ or temperature) has become limiting. NaHCO₃ provides a constant CO₂ supply.